University of British Columbia, Vancouver
Department of Computer Science

CPSC 304 Project Cover Page

Milestone #: 1

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Group Number: 13

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Project Description

Domain

The app is focused on financial markets and investment analytics. It is a stock market tracking and portfolio management database that stores information about public companies and their stocks, the people involved, and how market performance changes over different time periods. The goal is to support typical “market tracker” questions that offer meaningful comparisons over a period of time.

What aspects of the domain are modeled by the database?

The database models the core entities and relationships needed to track securities, participants, portfolios, and time-based market performance, including:

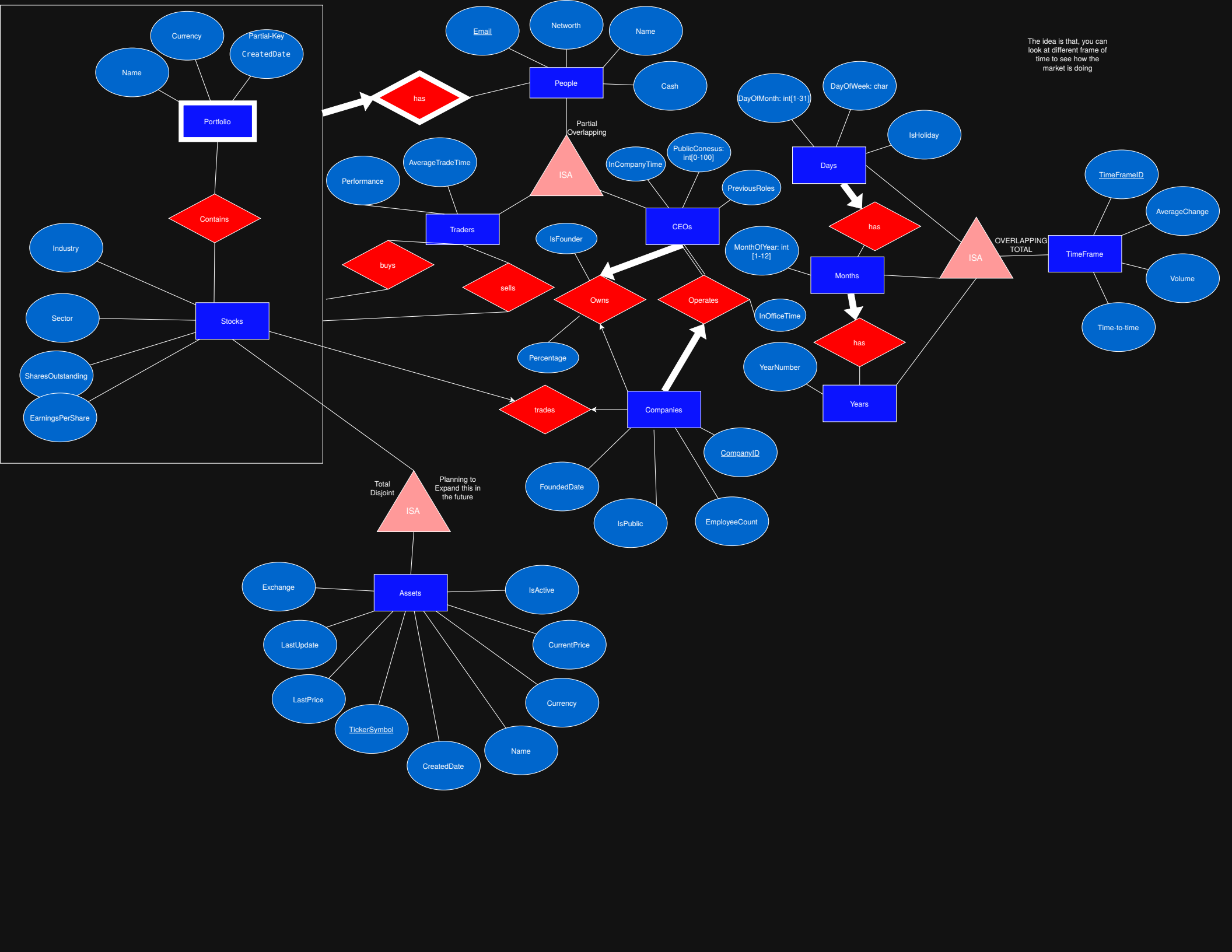
Market instruments and issuers:
The schema models companies and their associated stocks This captures the structure needed for stock tracking.

People in the market ecosystem:
A People entity specializes (ISA) into roles like Traders and CEOs, allowing the system to store shared person data (name/email/net worth/cash) while also supporting role-specific attributes (trader performance/average trade time, other important user attributes).

Trading activity and ownership relationships:
The schema includes relationships for buying and selling (traders buying/selling stocks), and links between companies/people such as a CEO operating a company and a person/entity owning something with a percentage. This supports real-world use cases like reconstructing activity histories and representing equity/portfolio allocations.

Portfolios and holdings:
Users can track portfolios (with attributes like name, currency, created date) and how they contain stocks, reflecting a real investment workflow.

Time-based performance tracking:
The schema models a TimeFrame concept with ISA categories such as Days, Months, and Years, and attaches market metrics like average change and volume over those time windows.


Currency
Partial-Key
CreatedDate
Name
Portfolio
Contains
Industry
Sector
SharesOutstanding
EarningsPerShare
Stocks
has
Email
Networth
Name
People
Cash
Partial Overlapping
ISA
Performance
AverageTradeTime
Traders
InCompanyTime
PublicConesus: int[0-100]
PreviousRoles
CEOs
buys
sells
IsFounder
Owns
Operates
InOfficeTime
Percentage
trades
Companies
CompanyID
FoundedDate
IsPublic
EmployeeCount
Total Disjoint
Planning to Expand this in the future
ISA
Assets
Exchange
IsActive
LastUpdate
CurrentPrice
LastPrice
Currency
TickerSymbol
Name
CreatedDate
DayOfMonth: int[1-31]
DayOfWeek: char
IsHoliday
Days
has
MonthOfYear: int [1-12]
Months
has
YearNumber
Years
ISA
OVERLAPPING TOTAL
TimeFrame
TimeFrameID
AverageChange
Volume
Time-to-time
The idea is that, you can look at different frame of time to see how the market is doing